Zhexiang (Bruce) Li

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Available for internships: Jun 13 - Sep 17, 2027

Education

University of California, Los Angeles Expected Jun 2028
M.S. in Computer Science

University of Southern California May 2026
B.S. in Computer Science; GPA: 3.76/4.00; Viterbi Dean's List

Research Skills

Methods: multimodal learning, efficient attention, token routing, molecular property prediction
Tools: Python, PyTorch, Triton, RDKit, Chemprop, C/C++, SQL, Linux

Research Experience

Graduate Researcher | Health Intelligence Lab, UCLA Aug 2026 - Present
Advisor: Prof. Yuzhe Yang

- Investigate multimodal glucose forecasting using CGM, insulin, meals, activity, and patient context.
- Built patient-disjoint MetaboNet benchmarks and data audits to compare GluFormer, T1DFormer, and custom models at 30-, 60-, and 120-minute horizons.

Drug Discovery Research | USC, Prof. Yan Liu's group Spring 2026
Research collaboration with a Ph.D. student in the group

- Benchmarked molecular descriptor-based models, Chemprop, and hybrid approaches across seven Therapeutics Data Commons datasets using scaffold splits.
- Analyzed descriptor importance and SHAP attributions to examine molecular features associated with predicted antiviral activity.

Selected Papers & Research Contributions

MedTokenBudget: Lesion-Preserving Token Routing 2026
Zhexiang Li (sole author). Preprint. Paper

- Developed lesion-supervised post-backbone token routing to retain diagnostic image regions under a limited token budget.
- At a 30% token budget on ISIC 2019, achieved **80.38 $\pm$ 0.18%** accuracy versus 76.93 $\pm$ 0.03% for ToMe across three seeds; the frozen backbone remains unchanged.

Mass-Conservation Token Transformers for MS/MS Retrieval 2026
Manuscript submitted to AAAI 2027

- Encoded mass-conserving fragment pairs as sparse evidence tokens for spectrum retrieval; constructed mass- and overlap-based hard negatives and evaluated ranking changes after evidence removal.

Efficient Attention via Pre-Scoring 2025
Zhexiang Li, Haoyu Wang, Yutong Bao, David Woodruff. Under review. Paper

- Proposed pre-scoring to prioritize informative keys before approximate attention; implemented key selection in Triton as first author, advised by Prof. David P. Woodruff (CMU).
- Integrated with HyperAttention, reduced ChatGLM perplexity from **12.0 to 9.5** at 131k-token contexts under a fixed interaction budget.

Teaching

Machine Learning Tutor | UCSC Spring 2023

- Led office hours and co-designed Jupyter assignments on logistic regression, SVMs, and tree models; contributed to the coding autograder for Prof. Lise Getoor's course.